DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103

Semester Examination – Dec - 2022

Branch: Electronics and Telecommunication Engineering **Sem.:- IV**

Subject with Subject Code: - Signals and Systems (BTEXC404) **Marks: 60**

Date:- **Time:- 3 Hr.**

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

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(Marks)

Q.1 A) Function $x(t)$ is as shown in fig. 1. Calculate and draw even and odd parts of the signal. (04)

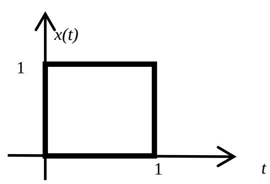

Fig. 1 : Given function $x(t)$

B) Compute the energy and power of the signal $x(n) = a^n u(n)$ where $|a| < 1$. (04)

C) 1. Check causality for the system given by equation: $y(n) = x(2n)$ (04)
2. Check linearity for the system given by equation: $y(n) = nx^2(n)$

Q.2 A) Find the convolution of a rectangle signal or gate function $x(t)$ as shown in fig. 2 with itself. (08)

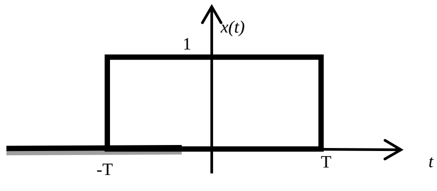

Fig. 2 : Given function $x(t)$

B) Function $x(t)$ is as shown in fig. 1. Find a) $x(2t-1)$ b) $2x(-t+2)$ (04)

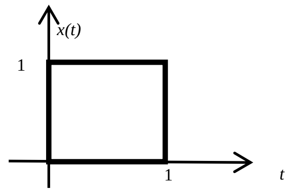

Fig. 3 : Given function $x(t)$

Q.3 A) If x(t) is an even signal, derive the formulas for trigonometric Fourier series coefficients a_0, a_n and b_n. (06)

B) Find exponential Fourier series for the waveform $x(t)$ shown in the fig. 4. (06)

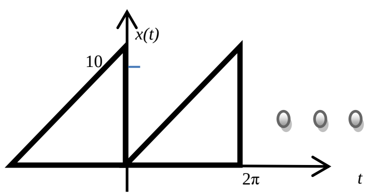

Fig. 4 : Given function $x(t)$

Q.4 A) Find ZT and ROC of (06)

1. $x(n) = \delta(n)$
2. $x(n) = \delta(n-k)$
3. $x(n) = a^n u(n)$
4. $x(n) = u(n) - u(n-10)$
5. $x(n) = a^n \sin(w_0 n) u(n)$

B) What is ROC in case of Laplace transform? State properties of ROC of Laplace transform (06)

Q.5 A) If output of an LTI system in response to an input $x(t) = e^{-2t}u(t)$ is $y(t) = e^{-t}u(t)$. Find the frequency response H(w) and impulse response $h(t)$ of this system. (06)

B) State and prove the following properties of the DTFT. (06)

1. linearity
2. time shifting
3. frequency shifting
4. time reversal
5. periodicity
6. differentiation in frequency domain

Q.6 A) Prove that the cross-correlation satisfies the following condition (06)

$$|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0)R_{yy}(0)} = \sqrt{E_x E_y}$$

B) State and explain Bayes' Theorem. Solve the following example with the help of Bayes' theorem. (06)

In a factory, four machines A_1, A_2, A_3 and A_4 produce 10%, 20%, 30% and 40% of the items, respectively. The percentage of defective items produced by them is 5%, 4%, 3% and 2%, respectively. An item selected at random is found to be defective. What is the probability that it was produced by the machine A_2?